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The Reinvention of Kodak

I don't feel pressure. I feel responsibility. Kodak is a century-old icon that deserves to fight another day.

— Jeff Clarke, CEO

In January 2016, Kodak CEO Jeff Clarke and his executive team traveled to Las Vegas, Nevada, for the annual Consumer Electronics Show (CES) where the company would unveil a prototype of its new Super 8 camera—an analog motion picture camera initially launched in 1965 to shoot home movies—with updated digital features. Over the course of the four-day event, the Kodak booth was overwhelmed by media and other industry players excited to catch a glimpse of the camera, asking when they could expect to see the Super 8 on shelves. To everyone's surprise, Kodak appeared to be back in the limelight.

The 133-year-old company dominated the film and photography industry through most of the 20th century. At its peak in 1997, Kodak had a market value of \$30 billion before entering into a period of decline as consumers eschewed analog film in favor of digital photography. Despite having invented the first digital camera, Kodak failed to capitalize on the new technology and by 2011 had entered into Chapter 11 bankruptcy. The “Kodak Moment”—a phrase that had signified the company's dominance for years—became a cliché to describe how companies should not respond to technological change.

In 2013, Kodak emerged from bankruptcy as a materials science company, operating in the commercial printing and packaging space, while still producing film. Clarke joined Kodak in March 2014 and was confronted with a series of key decisions to stabilize the business, which included assembling a new management team, cutting one-third of the workforce, and breaking up the company's business units into smaller reporting segments. As Clarke sought to identify opportunities for future growth, he questioned plans already in motion to shut down the legacy film business.

On the final night of CES, Clarke planned to meet with his team over dinner to discuss their annual progress. He wondered if the unexpected flurry of attention that the tech world had showered on Kodak's new Super 8 camera could be a symbolic turning point for the company's future growth plans. But many questions lingered. While the business-to-business (B2B) side of the company appeared to be growing, the new Super 8 reflected the culture and identity of a firm rooted in film and consumer products. How should a product like the Super 8 camera fit into the company's overall strategy? Was Kodak focusing on the right markets for growth? And should it continue to manufacture film?

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Kodak Origins

Kodak founder George Eastman was born in 1854 in Waterville, New York. In 1878, while working as a bank clerk, Eastman purchased a camera to take on vacation, as well as a tripod and the equipment necessary to develop photographs including chemicals, glass tanks, a heavy plate holder, a water jug, and a tent that he could develop photos without exposing them to light. Early cameras were heavy and expensive. Eastman described his equipment as a “pack-horse load.”¹ He became fascinated by photography and was determined to make a smaller, easy-to-use camera that was “as convenient as a pencil.”²

In 1880, in Rochester, New York, Eastman began producing dry gelatin plates, which replaced the bulky wet plates used in photography (see **Exhibit 1**). Wet plates had to be exposed and developed while still wet, but dry plates could be developed later, expanding photography’s reach, use, and accessibility. In 1888, Eastman founded Eastman Kodak Company (Kodak), and under the slogan “You press the button, we do the rest,” unveiled the company’s first camera, pre-loaded with a 100-exposure roll of film for \$25, equivalent to around \$600 in purchasing power in 2017.³ (See **Exhibit 2**.) Kodak was the first company to mass produce flexible film, replacing dry plates with 35 millimeter (35mm) film which quickly became the dominant film of the 20th century. By 1896, Kodak had produced a pocket-sized camera that sold for \$5, and in 1900 introduced the first of its long-running “Brownie” line of cameras for \$1.

Kodak’s color film, Kodachrome, was first used in movies in 1935 and for professional photography and print media. Kodachrome remained popular for most of the 20th century. By 1939, the company produced photographic paper, gelatin, and chemicals used to develop photos. By 1950, it operated several lines of business related to the production of film-based still and motion photography.

In 1963, Kodak unveiled its Instamatic 100 camera, and in 1965 released its Super 8 camera, a motion picture camera used to record home movies on 8mm film. By 1970, the company sold over 50 million units. (See **Exhibit 3** for product images over the years.) In 1975, a Kodak engineer developed the first working digital camera. The 8 lb., 0.01 megapixel camera took 20 seconds to capture an image, and needed to be hooked up to a television in order to view. In 1976, Kodak entered the instant photography market dominated by Polaroid, but was sued for patent infringement and forced out of the market in the 1980s.

Kodak grew tremendously into the 1980s. Its strategy was to sell cameras at an affordable price. Sales of film, chemicals used to develop photos, and photographic paper generated most revenues. High-profile ad campaigns encouraged consumers to preserve family events and milestones, which the company dubbed “Kodak moments.” Scientists at Kodak Research Labs developed some of the world’s leading innovations in photography. Chief Technology Officer (CTO) Terry Taber, who joined Kodak in 1980, recalled, “I came to Kodak because of the reputation of the research labs. There was a variety of broad-based, applied science being carried out on imaging technologies, alternative energy technology, and new smart materials. It was a very exciting place to be.”

Kodak owned much of the supply chain and manufactured its film and paper products in-house at Eastman Business Park, which grew into a 1,250-acre technology center and industrial complex 50% larger than New York’s Central Park. It contained 16 million square feet of manufacturing, lab, warehouse, and office space, and was served by a private railroad allowing for easy movement of materials and products in and out of the park. As the Park grew to 60,000 employees, work shifts were staggered in seven minute increments to manage traffic flow. Brad Kruchten, president of the print systems division, recalled, “When I joined Kodak in the 1980s as a quality engineer, we had very

defined tasks. 'Do this job, do it well, do it this way, and do it these hours.' When you have a large company, you need to just do your part, or there's chaos."

In Rochester, Kodak was known as "The Great Yellow Father" — a reference to its yellow film boxes. Kodak Scientific Photography and Marketing retiree Gordon P. Brown recalled, "The heyday was wonderful. The pay was good, and you had a job for life. My wife and I counted 45 blood relatives who worked for Kodak." The company had onsite a 14,000 square foot employee recreation center with bowling alleys and a shooting range, plus an onsite movie theater that provided free screenings. Kodak also hosted a dental clinic to provide affordable dental care to the Rochester community.

By the 1980s the industry was evolving, with international competitors beginning to encroach on Kodak's U.S. dominance; Japan's leading player, Fujifilm, aggressively undercut retail prices, and won sponsorship of the 1984 Los Angeles Olympics, catching the attention of U.S. consumers. By 1985, Kodak maintained 80% of the U.S. film market, and in 1988 had over 145,000 employees worldwide.

The Shift to Digital Imaging

Under CEO Kay R. Whitmore (1990-1993), Kodak invested heavily in its Photo CD product, a compact disc (CD) that stored up to 100 digital pictures. One analyst described it as the "natural bridge to a digital world of photography without film."⁴ Kodak announced software technologies for desktop publishing and printing, including scanners. The company also branched into pharmaceuticals.

In 1993, Kodak's board sought to bring down costs and named Motorola's George Fisher — widely credited with that company's success — the new CEO. By 1995, Kodak's U.S. market share in film had dropped to 70%⁵ as many retail partners switched to Fujifilm's lower-cost paper. More amateurs and professionals transitioned to digital photography, and Kodak struggled to produce less film without driving up unit costs.

In 1996, Kodak released its DC20 line of digital cameras. By then, the digital camera market was crowded with offerings from Canon, Hewlett-Packard (HP), Sony and other consumer electronics manufacturers. Kodak sales peaked at \$16 billion with profits of \$1.3 billion, but in 1997 the digital business was "hemorrhaging cash," — and profits fell precipitously to \$5 million.⁶ In response, Kodak announced a restructuring and cut 19,000 jobs. (See **Exhibit 4** for historical share price and number of employees.) By the late 1990s, a growing number of movies were shot digitally. Kodak did invest in digital imaging, over time accruing more than 1,000 digital imaging patents⁷, but the company's core competencies remained in chemistry and film.

According to Willy Shih, former Kodak SVP and head of the consumer digital business, the heavily-integrated company struggled to pivot: "Several factors made it exceedingly difficult for Kodak to shift gears [to digital]. [. . .]," said Shih. "Digital imaging was based on a general-purpose semiconductor technology platform that had nothing to do with film manufacturing."⁸

Long-time Kodak employee Daniel Carp succeeded Fisher as CEO in 2000. In 2001, Kodak released its EasyShare line of digital cameras. Digital camera sales first exceeded sales of film cameras in the U.S. in 2003, and globally by 2005.⁹ In 2004, Canon and Sony led the global digital camera market with 17.1% and 16.7% market share, respectively, while Kodak held 11.8%.¹⁰ That year, Kodak announced it would stop selling 35mm film cameras in the U.S., Canada, and Europe. Meanwhile, Kodak tried to make inroads in the inkjet printing business to compete with HP, Canon, and Epson and spent hundreds of millions to acquire print-related businesses NexPress and Scitex.

In 2005, HP executive Antonio Perez replaced Carp as CEO. That year, Kodak was the top seller of digital cameras in the U.S., overtaking Sony, and for the first time generated more in sales from digital imaging than film-based photography. “Management has been criticized for compromising its digital efforts because it wanted to protect film. But the criticism is overblown,” Shih noted.¹¹ In 2007, Kodak enjoyed its first year of profitability since 2004. “But then the financial crisis hit...,” Tim Palmer, vice president development, Eastman Business Park and corporate real estate, recalled. The film business, meanwhile, was “falling too precipitously for the company to gain overall sales or profit momentum.”¹² In 2008, Kodak’s traditional films and paper business represented only one third of its revenue versus 70% just a few years prior. Kodak’s stock price dropped from \$23.62 in 2006 to \$6.08 in 2008.

Adjusting to the Digital Decline

By 2008, the digital camera market was in decline as new smartphones with built-in, such as the Apple iPhone, were introduced. At that time, over 120 million camera phones were in circulation in the U.S. alone.¹³ The digital camera was “essentially decimated,” one expert noted.¹⁴ (See **Exhibit 5** for smartphone growth and digital camera decline.)

Between 2009 and 2010, Perez worked to reposition Kodak as a powerhouse of commercial printing, offering inkjet and packaging printing, and printing-related processes for manufacturing.¹⁵ Kodak was short on cash given declining camera sales, exacerbated by the economic downturn. In September 2011, the company announced it was taking \$160 million from a \$400 million credit line, alarming investors.¹⁶ Kodak’s share price fell more than 65% in 2011, trading for less than \$2. Analysts expected Kodak’s commercial printing business to start generating profits from new products by 2013, but as one wrote, Kodak needed to “stop the bleeding.”¹⁷ Rival Fujifilm had fared better, as it was quicker to cut costs and diversify. Fujifilm used its library of chemicals to produce antioxidants for the cosmetics industry as well as optical films for LCD flat-panel screens.¹⁸

Bankruptcy

In 2012, Kodak, having failed to post profits since 2007, filed for bankruptcy with \$5.1 billion in assets and nearly \$6.8 billion in debts. “We were struggling under the weight of pension liabilities and a capital structure that didn’t really work,” Chief Financial Officer (CFO) David Bullwinkle recalled. “We were investing so much in the growth opportunity in printing, which needed to be funded through other businesses.” Ed Hurley, general manager of manufacturing external sales who had been with Kodak for 36 years, noted, “When the company went into bankruptcy, we had serious cash issues. Although we had great technologies and ideas for what we could do with our assets, we really just didn’t have the capital. We had to let a lot of really good, capable people go, just to survive.”

By 2012, Kodak reduced its workforce to 17,000 employees, closing 13 manufacturing plants and 130 processing labs. Kodak sold off 1,100 digital imaging patents and exited legacy businesses. It further downsized to 9,000 employees. Perez oversaw bankruptcy and agreed to continue as CEO for one year after Kodak emerged from Chapter 11. Perez helped with the CEO search and planned to work with the new CEO to ensure a smooth transition.

In September 2013, Kodak emerged from bankruptcy as a smaller digital imaging company. Perez had repositioned it as a printing technology company, specializing in packaging and printing-related technology to manufacture products such as touchscreen thin films. Kodak leadership believed the firm was finally on a trajectory for profitable growth. Still, Kodak struggled to reestablish its identity and credibility, externally and internally.

A New CEO: Jeff Clarke

In March 2014, Clarke was named CEO. Clarke, a native of upstate New York and graduate of SUNY Geneseo, had three decades of experience in the technology sector. In 1985, he joined Digital Equipment Corp. (DEC), serving in various management roles. He joined Compaq in 1998 when it acquired DEC, and in 2001 became CFO. Clarke co-lead the merger of HP and Compaq in 2001, and then served as EVP of global operations. After two years as chief operating officer at CA Technologies, he became president and CEO of Cendant travel services. In 2006, Clarke was named CEO of Travelport (before it was acquired by the Blackstone Group), and managed the 2007 initial public offering (IPO) of Orbitz, owned by Travelport. In 2012, he co-founded private investment firm Augusta Columbia Capital. Kodak Board Chairman James V. Continenza stated, "Jeff is the right person to lead Kodak forward. His combination of strengths and experience in technology, transformation, finance, operations, and international business is precisely what we set out to find in the next leader of Kodak."¹⁹

When Clark first visited the 19-story Kodak Tower, he encountered many portraits of George Eastman on display as a tribute to Kodak's past dominance. Elevators bound for the executive offices on the top floor—which had a security guard on duty during past CEOs—had been programmed to not stop at other floors. Clarke observed that many of those floors were now dark with empty cubicles.

"Having grown up in upstate New York where Kodak was the largest employer and the most storied company in the region for years, the opportunity to come back and help lead Kodak was a very interesting opportunity," Clarke explained. "I've been in technology and software for most of my career, so to go into a materials science company was a bit surprising for many." Louise Kehoe, a former Kodak VP and Director of Media Relations, explained, "I've known Jeff for years, and when he told me he was joining Kodak, I was quite surprised. He's a technology guy, a Silicon Valley guy. But he was interested in the challenge of bringing back this iconic brand and iconic company."

"We were excited about Jeff. There was a lot of excitement about our growth prospects, and a lot of new technologies being developed in 2014," Bullwinkle recalled. "There were certainly high expectations. Jeff has had a very impressive career and has done tremendous things at Compaq and HP, so I was really excited to get him in the door and learn from him."

Immediate Decisions: Resuscitating the Crippled Yellow Giant

When Clarke joined Kodak, he estimated about 90% of the business was in decline. "We had to first cut costs, close factories in businesses that were declining, and generate cash so we could invest in the 10% of Kodak that was growing." Despite Kodak's struggles over the years, Clarke observed a strong company culture. "This was the first company to really embrace heavy advertising and simplicity of product. We wanted to keep that part of the culture." Clarke wondered, however, "How do you maintain that culture in a business that is declining 90%, needs to eliminate 30% to 40% of positions, and infuse new, Silicon Valley-like traits? I had to move incredibly fast and make a series of decisions right away. Bankruptcy doesn't solve all problems, it just gave us a fresh start."

Clarke described his leadership style as one centered on data-driven decisions and complete transparency. "I have encouraged my management team to bring me bad news. I encourage it and reward it, so we can get it on the table and fix it. You can't run a turnaround and not find bad news." Kruchten added, "Jeff's background is in startups, so he wanted us to move from a slow, methodical structure, to a company that moved faster. He wondered how he could make Kodak operate more like a startup instead of a 120-year-old company."

Clarke first met with each board member individually to discuss the company's challenges. He recalled, "The board believed there were certain technologies that were going to be transformative for the company going forward, and certain businesses in decline that needed to be managed for cash to fund new investments. At a high level, there was a simple recipe of how we would fix Kodak through commercialization of technology that was coming from the research labs." Clarke continued, "Kodak was in a position where its legacy costs were behind it, and there was a huge opportunity for Kodak with functional printing and printing on packaging."

Assembling a Team

Clarke assembled his core leadership team from a mix of new hires and Kodak employees who he believed had excelled under difficult circumstances. In 2009, Taber was named CTO and remained in that role under Clarke. Bullwinkle, who first joined Kodak 2004 as part of the controller's organization, served as director, corporate financial planning and analysis and vice president of the corporate finance group, and was later named CFO in 2016. Kruchten joined Kodak as a quality engineer in 1982 and by 2012 was named president of the Graphics, Entertainment and Commercial Films segment. In 2015 he was named head of the newly-formed Print Systems unit.

Clarke hired Mark Green, who previously served as VP of human resources (HR) at semiconductor supplier Applied Materials, as chief HR officer. "I knew Jeff for a while, and he called to say he was taking the job at Kodak and he needed someone that could come in and hit the ground running." He continued, "This is the classic example of a big turnaround, and as an HR person a great opportunity." Clarke also recruited Steven Overman, former VP and global head of brand strategy and marketing at Nokia, as chief marketing officer (CMO). "There was no question that I would want to be part of this. It was incredibly easy to recruit me to Kodak," Overman explained.

Clarke kept his primary residence in San Francisco and regularly traveled to Kodak headquarters in Rochester. Many members of his leadership team were based globally. "In the past, almost everyone had to live in Rochester. Now, my head of business development is in San Francisco, head of software is in Vancouver, head of Ink Jet is in Dayton, Ohio [. . .]. We're a global company," Clarke explained. Overman, based in London, added, "My team is largely virtual. My direct reports are in London, Rochester, California. I sometimes think it would be easier if we were one culture in one place, but there's something very contemporary about figuring out how to navigate a different way of working."

The leadership team held meetings in a different city each quarter. Overman added, "Jeff's trying to force a mindset that it isn't just about what happens in Rochester, but in the marketplace. The headquarters is a corporate ruin in many ways, with floors of empty cubicles. I find myself wondering what it's like for people who work there every day, passing floors and cubicles where friends, colleagues, and rivals all once worked, and what that would feel like."

Downsizing

One of Clarke's first acts as CEO was to cut one-third of the workforce. "We didn't have enormous runway, so the first thing we did was a very aggressive cost cut to stabilize the business," Clarke explained. Green added, "We couldn't afford to keep 9,000 people. We also had to make difficult adjustments to compensation and benefits programs, including pensions. It was a big undertaking." To reduce costs, Kodak offered an enhanced severance package to some employees and introduced a voluntary workforce transition program to others to continue working part-time to reduce costs. Among Kodak's 60,000 retirees in the U.S., 30,000 lived in Rochester, and 2,000 out of Kodak's 6,000 remaining employees were based in Rochester. "Nearly everyone in the Rochester area is touched in

some way by Kodak. We are so central to what happens here that we need to be very cognizant of our responsibilities,” Clarke explained.

In parallel to layoffs, Kodak asked employees to fill out an organizational health index survey (see **Exhibits 6a** and **6b**). “It was important to collect data,” Mark Green, chief HR officer, explained. “We asked employees questions such as, ‘What are the desired future values you’d like to see at Kodak?’ Responses included better employee focus, more professional growth, improved accountability, better execution, and trust.”

Breaking up the “Mega-Units”

In 2014, Kodak was operating two large business units: Graphics, Entertainment & Commercial Films (generating \$1.4 billion, or 68% of company sales), and Digital Printing & Enterprise and overlay geographic regions. “With two large units, it was hard for investors to understand the complexity in our portfolio,” Bullwinkle noted. Green added, “These ‘mega units’ were a combination of businesses with limited logic as to why certain products were in what unit, and unclear accountability between these BUs and the regions.” Reorganizing the business units, Clarke believed, would provide insight into the health of the business and help the team determine what parts of the business were worth investing in. Clarke also predicted that a new structure would allow the company to move faster and be more competitive, as well as sharpen its focus on accountability, performance, and predictability.

Clarke broke up the two units into seven: Print Systems, Enterprise Inkjet Systems, Micro 3D Printing and Packaging, Software and Solutions, Intellectual Property Solutions, Eastman Business Park, and Consumer and Film. The new structure went into effect January 1, 2015. (See **Exhibits 7a** and **7b** for more details.) “What holds all of us together is that we are a materials science company in service of creativity,” Overman explained.

Print Systems consisted of prepress solutions and electro photographic printing solutions. Kodak sold and serviced printers to 15,000 graphic arts and commercial print customers around the world, mostly small family-run businesses that printed newspapers, direct mail, and other material. “People buy from Kodak because they know it’s a quality product,” Kruchten explained. A key product was Kodak’s SONORA, a line of process-free printing plates, eliminated the energy, water, and chemicals required by processed plates without impacting quality. “Our ability to lead the print industry in a sustainable way is critical,” added Kruchten. SONORA was growing at 25% a year in the largely flat print market. Competitors in plate business included Agfa and Fuji, and in high-speed printing, HP and Xerox.

Enterprise Inkjet Systems had two lines of business: commercial inkjet printing solutions and digital front-end controllers. Kodak’s PROSPER 6000 printer, launched in 2014, was used for commercial print and publishing applications.

Micro 3D Printing and Packaging offered solutions for original equipment manufacturers (OEM) and major food and retail outlets using its KODAK FLEXCEL NX Systems and Plates. Kodak used a flexographic packaging process that provided improved color and resolution and produced more environmentally-friendly packaging. Christopher Payne, general manager of flexographic packaging solutions within the unit, explained, “P&G, Pepsi, and other consumer goods companies employ over half a million product managers. The Kodak brand is a calling card that enables us to open up a dialogue with these brands.” Kodak also provided materials used for 3D printing. Kodak competed with DuPont, 3D Systems, startups, and other materials science companies in this space.

Software and Solutions was comprised of Kodak technology solutions and unified workflow solutions including its KODAK PRINERGY Workflow Software.

Intellectual Property Solutions managed the Kodak Research Laboratories, licensing, and business development activities related to the company's patent portfolio and intellectual property. Taber, who was named CTO in 2009 and President, Intellectual Property Solutions in 2015, noted, "Kodak Research Labs has its own brand equity in the technology world. If I cold-call a company and say 'I'm CTO at Kodak,' that automatically opens doors."

Eastman Business Park was also named a separate business unit. In 2008, as film production dropped, Kodak began renting out space to other companies and shared a utility grid with tenants. Before Clarke joined, Kodak tried to sell the Eastman Business Park, but after receiving offers that management felt did not reflect the value of the Park, they decided to grow its value. In 2014, Eastman Business Park had 57 tenants consisting of companies in biopharma, chemical, food and beverages, plastics, and other big utility customers. Kodak Center, a movie theater and performing arts space, was also located at Eastman Business Park. Clarke and his leadership team considered whether Kodak should continue to grow the park and attract new tenants, or sell off this large fixed asset.

Consumer and Film included consumer inkjet solutions motion picture, industrial chemicals and films; and consumer products. The consumer inkjet business was declining 40% year-on-year, so by 2014 it was only selling replacement ink cartridges. Clarke sought to turn brand licensing, which had previously been an extra revenue generator collecting royalties from batteries, power cables, and other products, into a strategic force. Overman, named head of the division in January 2015, commented, "My marching order from Jeff was to triple the value of the brand."

Clarke believed that this new divisional structure would have a positive impact on Kodak's culture. Green agreed: "We have a strong company culture with high integrity and professionalism, but it was also complicated, slow, and there was a lack of accountability in making decisions. There was a lot of 'That's not my responsibility.' That's part of why we created these new more self-contained units." Louise Kehoe, VP, communications, added, "Each division has a fair amount of autonomy. This new divisional structure also brings transparency."

Eliminate Film?

From the early 2000s, movie producers had turned increasingly away from traditional film in favor of more flexible video and digital production. A handful of well-known Hollywood directors and producers strongly resisted this shift. In 2012, director Christopher Nolan told a reporter, "For the last 10 years, I've felt increasing pressure to stop shooting film and start shooting video, but I've never understood why. Film is far better looking. It's the technology that's been known and understood for a hundred years, and it's extremely reliable."²⁰ Director Quentin Tarantino insisted, "If I can't shoot on film I'll stop making movies."²¹ By 2014, film was experiencing a resurgence (see **Exhibits 8a** and **8b**).

The film business was profitable, but Clarke did not believe it had a lot of growth prospects. The leadership team considered how shuttering the film business could negatively impact the Kodak brand. Kurt Jaeckel, director of strategic worldwide communications, noted, "If you look at our communications, it's disproportionately focused on film and movies because people are interested in that. It's very hard to make offset plates sexy. That's something we're working on." Kehoe added, "In some parts of Kodak, film is a sacrosanct. It's very important to the history of the company and brand."

Initial Results

In 2015, Kodak reported revenues of \$1.8 billion, down 15% from the previous year (see **Exhibits 9a-d** for financials.) Kodak's largest division, Print Systems, represented over half of Kodak's revenues. In an earnings call, Clarke called it a "stable, predictable, and profitable business." Sales of Kodak SONORA plates, the unit's key growth product, had grown by 50%. SONORA had more than 200,000 customers worldwide, and opened a new manufacturing center in Georgia to meet growing demand. The Enterprise Inkjet Division enjoyed success with its lead growth product, the Kodak PROSPER printer. In 2015, Kodak released PROSPER 1000 Plus Press, the world's fastest black and white inkjet printer. One analyst commented, "We think this management team has redeemed itself to a certain extent in a tough operating environment, and we hope it is a start of a trend, of setting appropriate expectations and then delivering."

Kodak also reported strong growth in the Flexographic Packaging and Micro 3D Printing Packaging division. It installed FLEXCEL NX systems in Q2 2015, bringing the total units sold to more than 440. "Packaging continues to flourish, whether it be from Amazon or some other e-commerce company, or in retail," Clarke noted. Payne explained, "This division is small. We're less than 5% of the people at Kodak. But productivity per person is far higher than anywhere else. Our biggest struggle right now is how to grow the business fast enough."

"From a research standpoint, we have so many different technologies. One thing Jeff has had to do is try to choose where to make investments. We don't have unlimited budget, so it's tricky," Kruchten explained. Taber added, "We're working on so many cool things. For instance, we're working on particles that can control light, to integrate into our new products. We're forming partnerships with companies that can help us go to the market successfully." He continued, "Before Chapter 11, we surmised that functional printing, the ability to actually use printing technologies to do something beyond text and graphics, would be a foundational shift. We are starting to see that."

By 2015, Eastman Business Park had 60 tenants and was attracting new businesses to the park. "We have capacity here that is not being utilized. We're focused on attracting the right kind of companies that create value for themselves, for the park, and for Kodak by being here," Palmer explained.

In Consumer and Film, the consumer inkjet business continued to wind down, with Kodak just selling replacement ink cartridges.

Adapting to Changes

Many Kodak employees responded positively to the new direction of the company under Clarke. Megan Cucci, director of business and market research for the advanced materials & 3D printing technology division, explained, "Under the new leadership, we've thought about how to tackle technical and business challenges in a new way." Kruchten added, "Compared to when I first joined in the 1980s, we're now operating as a more entrepreneurial, start-up company," Kruchten noted.

"We're moving forward focused on what we're good at, and what we have been good at for years," Kevin Rhoney, PR and media relations specialist, explained. "The biggest misconceptions about Kodak come from the fact that people just don't see what we do. Before I joined this year, I really didn't know that Kodak was mainly focused on print and B2B activities." "It's challenging to tell the Kodak story," Kelly Mandarano, director of marketing and communications for print services, explained. "We have a set of legacy businesses as well as growth businesses. How do you tell the print story while telling the motion picture story to consumers?" Jaeckel added, "We talk about the elevator pitch for Kodak. Well, it's a long elevator ride because we have several different businesses within those divisions."

Clarke had earned the trust of many Kodak employees. Green explained, "Jeff has been a role model in transparency with the organization. The messages were tough but honest. It's only through that that you build trust." Palmer added, "Jeff does a lot of town halls with employees. He has a lot of credibility with people. He's a straight shooter. Not that he's not enthusiastic and positive, but he doesn't sugarcoat things, and people respect that."

"I think people are, in general, excited about where the company is going, happy to still be working here, and much more energized than they were three, four, five years ago," Hurley, who left Kodak in 2014 and returned in 2016, explained. "The mood at Eastman Business Park is much better than it was years ago. Morale has dramatically improved. Everybody knows the company is in the midst of this transformation. When I came back in 2016, it was like walking into a different company. There's now a kind of excitement about where we're going."

"One thing I've noticed is that we have a large fan base," Palmer observed. "A lot of people out there love Kodak the brand and Kodak the company. They're rooting for us to succeed. That's true of suppliers, customers, and so forth. So there's a lot of good will associated with the company." Some employees were attracted to Kodak to help rebuild the company. Evan Spinoso, regional business manager for the flexible packaging division, explained, "Knowing what had happened to the organization, having the chance to be part of its recovery was an opportunity I couldn't pass up."

After years of layoffs, the company was hiring again. "I don't think people realize how many hires Kodak made in the last few years," Hurley explained. "In our manufacturing division at Eastman Business Park we have about 800 employees, and about 100 of them have one year or less with the company. People think of Kodak as being a very mature workforce, and it's true, but there's a whole new group of people joining the company with fresh new ideas, and some just starting their careers."

Spinoso described the mood around headquarters: "Coming to work in the morning, you park in a half empty lot and walk past vacant offices piled with furniture, but just take the elevator to the 10th floor and we're bursting at the seams. Every office is filled, people are in early and working late. It's a stark contrast, and a powerful reminder of what's at stake." Mandarano added, "Kodak might never have 60,000 employees in Rochester again, but the park may be filled again. The parking lots may be filled again. It just might have a different face, and that benefits Rochester."

Excitement at CES

In 2015, Overman and Clarke began to brainstorm a new consumer product. "We needed to create a new product for multiple purposes," Overman said. "First, to prove we could. Second, to deliver innovation to the marketplace. And third, to iconize what Kodak would stand for going forward." Within six months, he learned of a family business in Denmark that was manufacturing cameras much like Kodak's original Super 8. "Something clicked," he said. He approached Clarke and suggested that Kodak re-release the Super 8 as an analog film camera, but with modern digital features. Clarke was immediately drawn to the idea. "We're pretty analytical and unemotional when we think about return on investment around our consumer products, but the Super 8 camera is where we probably deviate from that the most," Clarke explained.

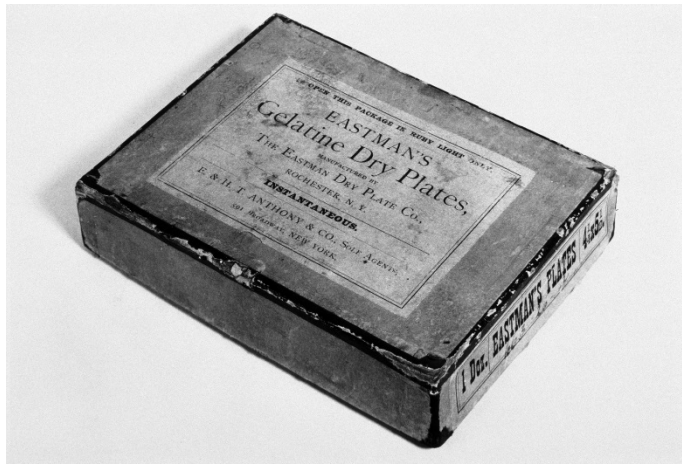
The following year, Kodak unveiled a reengineered prototype Super 8 camera at CES. Kodak's booth was immediately overwhelmed with press and representatives from other companies, all eager to catch a glimpse of the Super 8 prototype. "From the minute the doors opened until they closed four days later, the Kodak stand was just mobbed by the media," Kehoe added.

Clarke and his team were caught off guard by the buzz around the new product. Jaeckel noted, “People still have an emotional connection with Kodak. That’s hard to measure and can’t be bought. I think if we can harness that somehow, it’ll be very powerful for Kodak.” Kodak expected to release the Super 8 in the second half of 2016, priced between \$400 and \$750.²²

Positioned for Growth?

Back at the hotel, as Clarke headed towards the lobby to meet his team for dinner, he felt optimistic about Kodak’s future. The printing and packaging unit was building momentum, Kodak’s research labs were developing new technologies, and Eastman Business Park was attracting new tenants to Rochester. He was also excited about the release of the Super 8 camera. However, the company had not yet returned to profitability and continued operating on a tight budget.

As Clarke stepped onto the elevator, he was keenly aware of the expectations on him to turn around the iconic company. “My mandate from the board was very clear: to drive growth.” He considered Kodak’s culture and whether he had succeeded in instilling “Silicon Valley traits.” What elements of the business had the most potential? He considered the importance of film and the consumer-facing side of the business. Was film still an integral part of Kodak’s identity, or was it merely a distraction?

Exhibit 1 Kodak Dry Plates

Source: Kodak Dry Plates, Company website, <https://www.kodak.com/corp/aboutus/heritage/milestones/default.htm>, accessed October 2017.

Exhibit 2 Original Kodak Camera Advertisement, 1889

Source: Wikimedia Commons, [https://commons.wikimedia.org/wiki/File:You_press_the_button,_we_do_the_rest_\(Kodak\).jpg](https://commons.wikimedia.org/wiki/File:You_press_the_button,_we_do_the_rest_(Kodak).jpg), accessed October 2017.

Exhibit 3 Kodak Camera Images

Kodak Brownie Camera, 1896



Kodak EasyShare Digital Camera, 2001



Kodak Instamatic 100 Camera, 1963



Kodak PIXPRO Digital Camera, 2014



Kodak Star 110 c. 1990



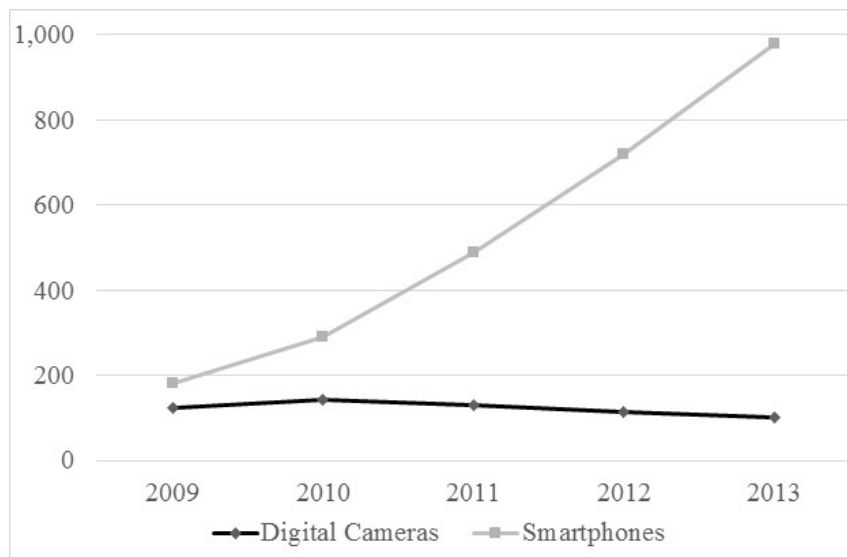
Cameo Motor EX Camera, 1995



Source: Company documents.

Exhibit 4 Kodak Share Price and Number of Employees, 1973-2013

Source: Compiled from Thomson Reuters Datastream; Standard & Poor's Compustat (via WRDS), accessed February 2018.

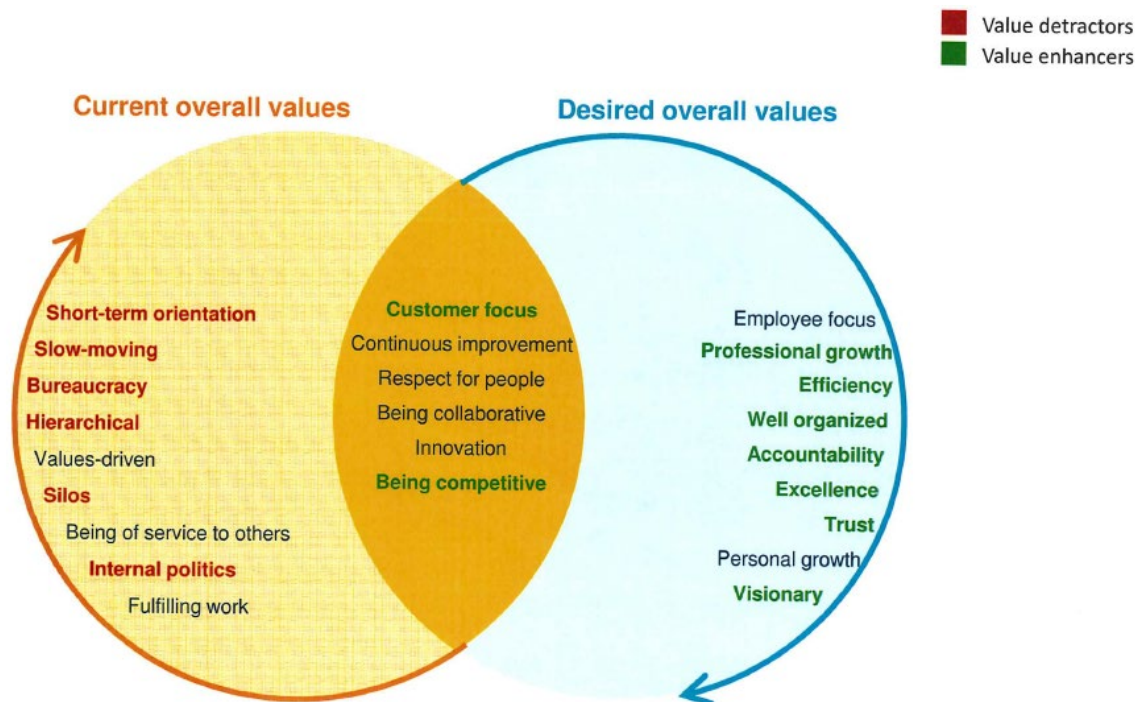
Exhibit 5 Digital Camera and Smartphones Unit Sales, Worldwide, 2009-2013 (in millions)

Source: Adapted from Daisuke Wakabayashi, "The Point-and-Shoot Camera Faces Its Existential Moment," July 30, 2013, *Wall Street Journal*, <https://www.wsj.com/articles/SB10001424127887324251504578580263719432252>, accessed January 2018.

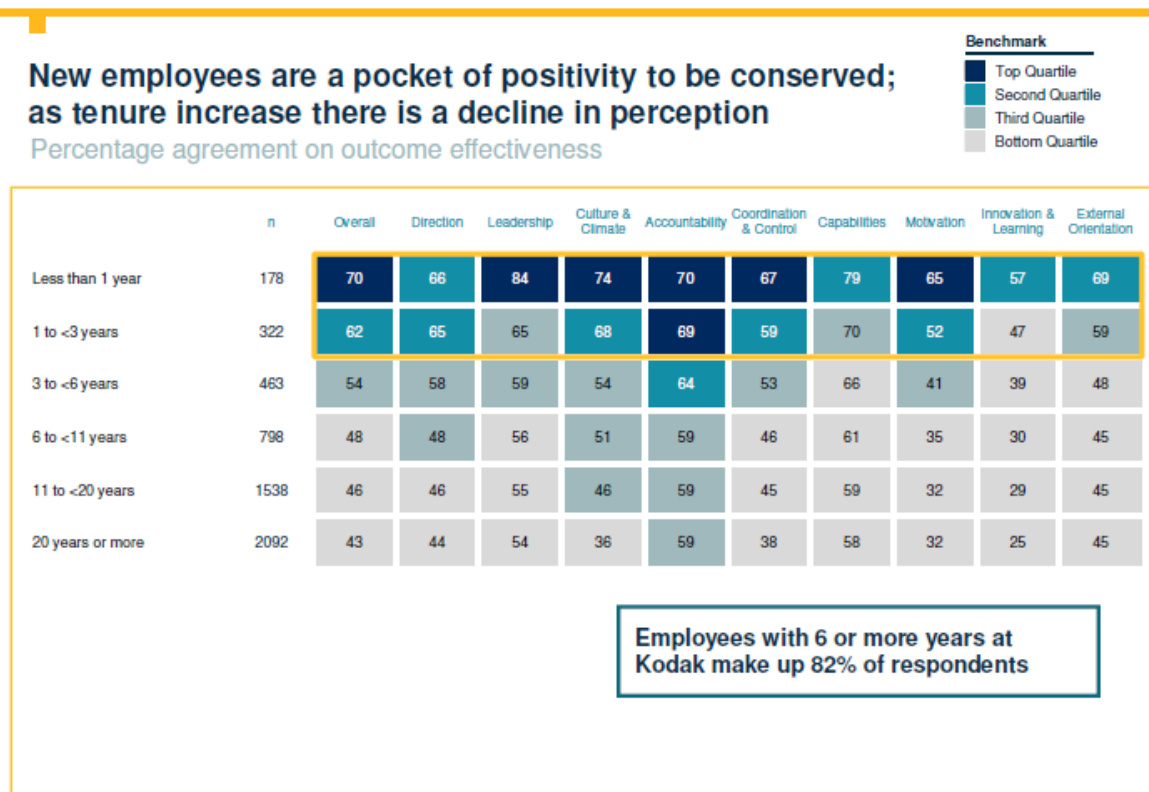
Exhibit 6a Kodak Organizational Health Index (OHI) Employee Survey Results, August 2014**KODAK PEOPLE WANT:**

- A clear vision and strategy
- An environment of innovation, creativity, and trust
- Investment in talent
- Visible leadership with transparency
- Differentiated rewards
- To play a role in the turnaround

KODAK VALUES: Creativity, Sustainability, Global, Trust.

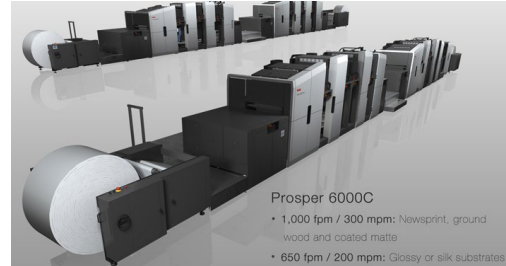
Top 15 Current and Desired Values

Survey response rate: 71%

Exhibit 6b Outcomes of Organizational Health that Drive Performance

Source: Company documents.

Note: This image shows the nine outcomes of organizational health that drive performance. For example, 84% of the new employees strongly agree/agree that leadership is effective.

Exhibit 7a Kodak Product Images by Division**Print Systems: SONORA****Enterprise Inkjet Systems: PROSPER 6000****Micro 3D Printing and Packaging: KODAK FLEXCEL NX Wide 5080 System****Packaged Printing Sample****Consumer and Film: Kodak Film****Super 8****SP4 Smartphone****Eastman Business Park**

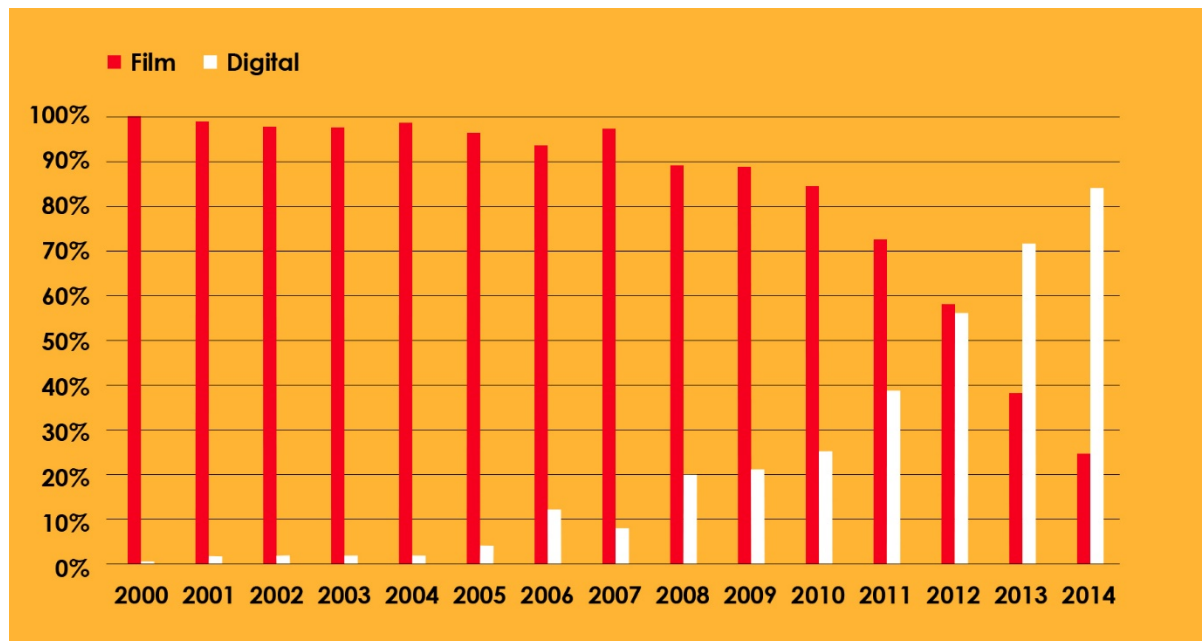
Source: Compiled from company documents.

Exhibit 7b Kodak Headcount by Division and Corporate Offices (2015)

Business Units	As of:	
	January 1	December 31
Print Systems	3,852	3,181
Consumer & Film	918	875
Enterprise Inkjet Systems	576	570
Software & Solutions	245	367
Micro 3D Printing and Packaging	216	252
Intellectual Property Solutions	218	176
Eastman Business Park	*	*
Corporate Offices	<u>1,121</u>	<u>927</u>
Grand Total	7,246	6,348

Source: Company documents.

* Embedded in one of the other organizations.

Exhibit 8a Shooting Format of the 100 Top-Grossing Films in the U.S., 2000-2014

Source: Adapted from "Film vs digital – What is Hollywood shooting on?" Stephen Follows Film Data and Education, January 11, 2016, <https://stephenfollows.com/film-vs-digital/>, accessed December 2017.

Exhibit 8b Academy Awards Best Picture Nominees Shot on Film, 2009-2014

Year	Nominated Films Shot Primarily* on Film
2008	4 of 5 nominees; winner (<i>No Country for Old Men</i>) shot on film
2009	3 of 5 nominees; winner (<i>Slumdog Millionaire</i>) shot on film and digital formats
2010	4 of 10 nominees; winner (<i>Hurt Locker</i>) shot on film
2011	3 of 10 nominees; winner (<i>The King's Speech</i>) shot digitally
2012	2 of 9 nominees; winner (<i>The Artist</i>) shot digitally
2013	3 of 9 nominees; winner (<i>Argo</i>) shot on film and digital formats
2014	3 of 9 nominees; winner (<i>12 Years a Slave</i>) shot on film

Source: Casewriter research, compiled from Internet Movie Database, <http://www.imdb.com>, accessed January 2018.

Note: Some movies have been produced using a combination of film and digital formats. Between five and 10 movies are nominated for Best Picture each year.

Exhibit 9a Kodak Consolidated Statement of Operations (in \$ millions, except EPS), 2013-2015

	2015	2014	2013
Revenues			
Sales	\$1,447	\$1,738	\$1,946
Services	351	378	412
Total net revenues	1,798	2,116	2,358
Cost of revenues			
Sales	1,171	1,376	1,545
Services	246	284	325
Total cost of revenues	1,417	1,660	1,870
Gross Profit	381	456	488
Selling, general, and admin expenses	226	310	411
R&D costs	61	94	99
Restructuring costs and other	38	59	60
Other operating expense (income), net	2	9	(493)
Earnings (loss) from continuing operations before interest expense, loss on early extinguishment of debt, net, other (charges) income, net, reorganizational items, net and income taxes	54	(16)	411
Interest expense	63	62	128
Loss on early extinguishment of debt, net	-	-	8
Other (charges) income, net	(21)	(21)	(3)
Reorganization items, net	5	13	(2,010)
(Loss) earnings from continuing operations before income taxes	(35)	(112)	2,282
Provision for income taxes	32	10	163
(Loss) earnings from continuing operations	(67)	(122)	2,119
(Loss) earnings from discontinued operations, net of income taxes	(8)	4	(131)
NET (LOSS) EARNINGS	(75)	(118)	1,988
Less: Net income attributable to noncontrolling interests	5	5	3
NET (LOSS) EARNINGS ATTRIBUTABLE TO EASTMAN KODAK COMPANY	(\$80)	(\$123)	\$1,985
Basic and diluted (loss) earnings per share attributable to Eastman Kodak Company common shareholders			
Continuing operations	(\$1.72)	(\$3.05)	\$6.04
Discontinued operations	(0.19)	0.10	(0.40)
Total	(\$1.91)	(\$2.95)	\$5.64
Number of common shares used in basic and diluted (loss) earnings per share	41.90	41.70	314.40
NET (LOSS) EARNINGS	(\$75)	(\$118)	\$1,988
Less: net income attributable to noncontrolling interests	5	5	3
Net (loss) earnings attributable to Eastman Kodak Company	(80)	(123)	1,985
Other comprehensive (loss) income, net:	-	-	-
Currency translation adjustments	(35)	(33)	5
Reclassification of realized losses on available-for-sale securities included in net earnings, net of tax	2	-	0
Pension and other postretirement benefit plan obligation activity, net of tax	(98)	(202)	1,702
Other comprehensive (loss) income, net attributable to Eastman Kodak Company	(131)	(235)	1,707
COMPREHENSIVE (LOSS) INCOME, NET ATTRIBUTABLE TO EASTMAN KODAK COMPANY	(\$211)	(\$358)	\$3,692

Source: Kodak 2015 Form 10-K and Annual Report.

Exhibit 9b Kodak Balance Sheet (in \$ millions), 2014-2015

	2015	2014
ASSETS		
Cash and cash equivalents	\$ 547	\$ 712
Receivables, net	365	414
Inventories, net	314	349
Deferred income taxes	22	31
Other current assets	30	44
Total current assets	1,278	1,550
Property, plant and equipment, net of accumulated depreciation	426	524
Goodwill	88	96
Intangible assets, net	158	182
Restricted cash	43	37
Deferred income taxes	23	38
Other long-term assets	122	129
TOTAL ASSETS	\$ 2,138	\$ 2,556
LIABILITIES AND EQUITY		
Accounts payable, trade	\$ 195	\$ 212
Other current liabilities	264	387
Total current liabilities	459	599
Long-term debt, net of current portion	675	672
Pension and other postretirement liabilities	623	662
Other long-term liabilities	278	324
Total liabilities	2,035	2,257
Equity		
Common stock, \$0.01 par value	n/a	n/a
Additional paid in capital	633	621
Treasury stock, at cost	(5)	(4)
Accumulated deficit	(283)	(204)
Accumulated other comprehensive (loss) revenue	(267)	(136)
Total Eastman Kodak Company shareholders' equity	78	277
Noncontrolling interests	25	22
Total equity	103	299
TOTAL LIABILITIES AND EQUITY	\$ 2,138	\$ 2,556

Source: Kodak 2015 Annual Report.

Exhibit 9c Revenue by Business Segment (in \$ millions), 2013-2015

	2015	2014	2013
Print Systems	\$ 1,106	\$ 1,257	\$ 485
Enterprise Inkjet Systems	173	185	83
Micro 3D Printing and Packaging	128	130	42
Software and Solutions	112	108	39
Consumer and Film	265	352	147
Intellectual Property Solutions	1	70	9
Eastman Business Park	13	14	4
All Other	n/a	n/a	3
Consolidated total	<u>\$ 1,798</u>	<u>\$ 2,116</u>	<u>\$ 812</u>
EBIDTA by Segment:			
Print Systems	\$ 98	\$ 93	\$ 38
Enterprise Inkjet Systems	(26)	(44)	(15)
Micro 3D Printing and Packaging	9	(1)	-
Software and Solutions	9	3	(1)
Consumer and Film	52	66	34
Intellectual Property Solutions	(22)	40	(2)
Eastman Business Park	2	1	4
Total of reportable segments	122	158	58
All Other	5	5	4
Depreciation and amortization	(145)	(199)	(75)
Corporate components of pension and OPEB income	133	110	67
Restructuring costs and other	(38)	(59)	(17)
Stock-based compensation	(18)	(8)	(1)
Change in U.S. vacation benefits	17	-	-
Consulting and other costs	(13)	(6)	(2)
Idle costs	(3)	(4)	-
Costs previously allocated to discontinued operations	(1)	(4)	(5)
Fresh start adjustments	-	-	(73)
Other operating (expense) income, net excluding gain related to Unipixel termination	(5)	(9)	(2)
Loss on early extinguishment of debt, net	n/a	n/a	n/a
Interest expense	(63)	(62)	(22)
Other charges, net	(21)	(21)	10
Reorganization items, net	(5)	(13)	(16)
Consolidated (loss) earnings from cont. ops. before income taxes	<u>\$ (35)</u>	<u>\$ (112)</u>	<u>\$ (74)</u>

Source: Kodak 2015 Annual Report.

Endnotes

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- ¹⁴ Shih, "The Real Lessons From Kodak's Decline."
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- ¹⁹ Matthew Daneman, "New Kodak CEO Jeff Clarke to focus on growth," *Democrat and Chronicle*, March 12, 2014, <http://www.democratandchronicle.com/story/money/business/2014/03/12/jeff-clarke-kodak-ceo/6318457/>.
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- ²¹ Matt Patches, "Theater owners decry 'Interstellar' 35mm rollout as Tarantino calls them 'barbarians,'" *Uproxx*, October 3, 2014, <http://uproxx.com/hitfix/theater-owners-decry-christopher-nolans-interstellar-35mm-rollout-quentin-tarantino-calls-them-barbarians/#theater-owners-decry-interstellar-35mm-rollout-as-tarantino-calls-them-barbarians/>, accessed October 2017.
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